KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI.



COLLEGE OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

AUTOMATIC CONTROL GROUP ONE (MARINE ENGINEERING)

GROUP REPORT.

TOPIC: Using a Programmable Logic Controller (PLC) to solve Valve issues onboard a Marine Vessel.

20th July, 2023

DECLARATION

We hereby declare that with exception of the references, diagrams and secondary data used, the content of this report was based on our individual inputs, through research.

ACKNOWLEDGEMENT.

We are grateful to Almighty God for giving us the strength, knowledge and understanding to complete this project. His love has been more than sufficient to keep and sustain us. We also extend gratitude and appreciation to our lecturer Dr Eunice Adjei providing us with the chance to work on this wonderful project, “Using Programmable Logic Controllers to solve problems associated with valves on marine vessels.” which also aided us in conducting extensive study and learning about numerous new topics. We are grateful to her all for her valuable guidance and suggestions all throughout the project duration. We would want to thank Ing. John Kusi for his massive support during the practical session of the project. We say Ayekoo! We also thank all friends and colleagues for their assistance in completing this project in such a short period of time. God bless you all

Date: 20th July, 2023

NAME	INDEX	NO SIGNATURE
ISAAC BLANKSON	8310819	……
AMOAKO BERNARD KOFI	8310519	……
EDEM KOMASHIE	8311219	……
ADDO DANKWAH EDEM	8309819	……
AKUMANE JOSHUA	8310319	……
ACHEAMPONG ANDY KOFI	8309719	……
ISAIAH LAMPTEY ODARTEI	8311519	……

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ABSTRACT.

Valves are mechanical devices that regulate, direct, or control the flow of a fluid (gases, liquids, fluidized solids, or slurries) by opening, closing, or partially obstructing various passageways. These are essential for fluid power systems. Modern control valves may regulate pressure or flow downstream and operate on sophisticated automation systems. They are widely recognized in the transport sectors delving deep into the automobile, marine, aviation, and railway sectors in as much as they are highly esteemed and recognized in top-notch industries that facilitate plant processing and automation. In the maritime sector, marine vehicles which include ships, submarines, and many other sea□going vessels over the years have been designed and well equipped with different types of valves designed to perform specific functions according to the purported or specified design. Ships need valves to regulate the pressure and flow of fluids in the various onboard systems. Along with liquids, valves can also transfer gases, vapors, and slurries or semi-liquid mixtures. Valves may be small, but they serve a big purpose in ensuring smooth sailing for Naval vessels and other ships. Programmable Logic Controllers have existed over the years and possess track records of being efficient in system controls and automation, especially in ships. These devices are stationed in the engine control room and the bridge of the vessel. A PLC is designed to check the input status, execute the program, and update the output status and can be used in system control of various components such as valve operations and monitoring as well as assisting in troubleshooting faults.

Keywords: Programmable Logic Controllers, Valves, Marine Vessels, Automation and Control Engineering.

CHAPTER ONE

INTRODUCTION.

1.0. Problem Identification.

Valves on board marine vessels are serviced periodically to ensure they are in good working conditions under the Preventive maintenance approach. Compared to any mechanical component, valves are prone to have faults that need to be troubleshot immediately as they contribute efficiently to regulating the pressure of fluid flowing through pipes from one place to another to avoid the likelihood of explosions should there be any pressure accumulation within the tanks. In today's practice of maintenance engineering as modern technology has evolved, troubleshooting a valve is done either manually or using specially aided- softwares to rectify these issues. These programmable softwares such as the PLC help reduce human efforts involved in troubleshooting valves for marine engineers onboard a vessel. During a class session on system representation analysis, our facilitator recognized the need of using a software to resolve these challenges and then instructed us to identify all problems associated with valves on board a vessel and use PLC programming to rectify these issues

1.1. What are Valves?

Valves play a very critical role in the marine industry. As ships becomes larger and are used more frequently, they require the ability to generate power, treat and manage wastewater and control HVAC, as well as perform their required tasks. The size and application of the ship will determine the different types and number of valves on board. Valves are used in almost all the machinery systems on the ships. Valves are used to regulate the loading and the storage of ships power supply, provide water for firefighting capabilities, handle and processes wastewater and store any liquid cargo, among many other applications. Any valve that processes sea water must be durable, and all marine valves must be reliable due to lack to resources once out at sea. Marine valves are recommended to be overhauled at a regular interval. This is because if the valve is always in contact with the sea water, with time, the seat, spindle, and other parts will lose it original state. Crating leaks and hard openings. Some valves are sealed with O rings on the spindle end, whiles some are sealed with gland packing. The gland packing or O ring can be renewed if the valve got leaked. There are four main functions of valve Regulate, Isolate, special purpose and non-return. All ships have some type of valve systems to control the flow of fresh water, ballast water, pumping out of bilges, transferring of bunkers and for pressurizing the hydraulic systems for various control operations. They come in all sizes and shapes and can be operated by almost all energy sources available on the ship including, manual labor. On tankers they assume even more importance as they help orderly loading and discharge of cargoes. Close the bypass valve before switching on the power for the system. The bypass valve connects the discharge line

from the pump, to the return line from the deck and when opened it allows the oil under pressure in the system to return to the reservoir. This valve should be kept closed when the system is operational and be opened only if there is a leak in the system whereby it becomes necessary to reduce the hydraulic pressure immediately. When a leak is detected, the pumps should be stopped, by pass valve opened and the pressure in the system is released. When switching on the power, the operator decides which pump is to be primary and which standby. This is done so that both the pumps are used alternately and one pump does not get worn out due to continuous use while the other suffers due to non-use. It is a good practice to check the level of oil in the reservoir regularly and a record should be maintained of any replenishment done after the system has started. It should be noted that any replenishment should be done with the knowledge of the chief officer and the replenishment should bring the level of oil to the level at which the pumps were started. If the reservoir is topped up while the system is in operation, the reservoir will overflow after the oil returns to the reservoir if it is already full. If the oil level falls during valve operation, then such valve should be identified so that operation of that valve could be avoided and the hydraulic line leading to the valve attended to at the first possible opportunity. The spare hydraulic oil drums should be marked and only the correct grade of oil kept in the hydraulic room to replenish the reservoir. When the switch on the control panel for a particular valve is operated, it sends a signal to the solenoid on deck, which operates to allow the oil to flow in the appropriate direction to operate the valve. 3 In case of emergency, the solenoid has an arrangement to connect a manual hydraulic pump to operate the valve in case of failure of the hydraulic system. The isolation valve on the hydraulic box housing the solenoid should be shut so that the pressure is applied within the lines in the affected area. Adjust the speed at which the valve operates.

1.2. Types of Valves used on board vessel.

There are four classified valve types designated for marine utilization; isolation valves, regulating valves, non-return valves and specialized purposed valves. They work hand in hand to provide an operation free from obstruction and potential downtime. It is well recommended that valves are used according to its function to avoid back pressure to the pump.

Isolation valves; An isolation valve is used in fluid management to stop the flow of process fluids in a pipeline, this is usually for maintenance or safety purposes. They can also be automated to provide flow path selection to change the flow path in a continuous or batch process.

Valves which can be found in this category are;

- Piston valves
- Gate valves
- Ball valves
- Diaphragm valves
- Butterfly valves

Regulating valves

Regulating valves are used for pressure sustaining, reducing and relieving valves to provide safe and efficient operating conditions and automatically respond to changes to maintain the set pressure. Valves which can be found in this category are:

- Globe valves
- Needle valves
- Control valves
- Pinch valves

Non-Return valves. Allow fluid to flow in one direction only, therefore preventing the liquid or gas from flowing back upstream of the valve. Example is the;

- Check valves

Specialized purpose valves.

Designed and manufactured to meet the challenges of complex piping systems. If a process or mechanical system requires an uncommon material or a very precise operating parameters that standard valves cannot meet, a special purpose valve can be designed to meet exacting specifications. Valves which can be found in this category are;

- Float valves
- Knife gate valves
- Line blind valves
- Multi-port valves
- Flush bottom valves
- Foot valves

1.3. Location of the various valve types on board the vessel.

Valves can be located on various equipment, machinery or marine systems in the ship especially in the engine room and other compartments on the ship. A variety of valves are used in the piping and machinery systems of the ship as per the need of the flow pattern of the liquid. The following are some of the locations of valves onboard a ship and these include: Engine room, Galley, Forecastle, Bridge, Accommodation, Pump room, Storage tanks, Steering room, and many other places just to name a few

CHAPTER TWO

CONTROL SYSTEMS

2.0. Valves control systems on board the ship.

These are valves that controls the fluid flow by varying the size of the flow passage as directed by signal from a controller. This enables the direct control of flow rate and the consequential control 9 process quantities such as pressure, temperature and liquid level. A control valve controller is also referred to as valve pilot or pressure pilot. A pilot used to monitor and control the upstream or the downstream pressure. It does this by positioning a control valve to achieve the producer's desired set point. The pilot sense a process condition typically pressure. It then sends a corresponding pressure signal to the control valve diaphragm. The diaphragm responds to that pressure by opening or closing the valve stem to regulate the flow through the valve. There are different types of controllers for valves, we have pneumatic pilot, and electrically actuated control valves use an electric pilot, Programmable logic controller PLC, or remote Terminal unit bra RTU.

The actuator is the device connected to the valve through the valve stem that provides the force required to move the valve. As we said earlier, the actuator can be controlled electrically, pneumatically, or hydraulically.
The most common and the most reliable is the "Pneumatic Actuator".
The control valve receives a signal from a controller such as a PLC or a DCS in order to operate. The controller compares the actual flow rate to the desired flow value called the set point. The controller will produce an output to move the valve to bring the flow rate to the set point value.
Because of how pneumatic actuators are built, a control valve will fail to a specific position when a loss of the control signal occurs. As the supply air pressure is increased, the rubber diaphragm pushes against the spring and moves the valve stem down into the valve body. As the supply air pressure is reduced, the spring will move the valve stem out of the body.
The position to which the control valve is moved if a loss of signal occurs is referred to as the fail□safe mode.
The type of fail-safe mode depends on the application for which the control valve is used. The actuator causes the valve to close in a Fail-Closed Control Valve. The actuator causes the valve to open in a Fail-Open Control Valve.
In normal operation, the force of the spring must be overcome by the electrical or pneumatic actuator. In the event of a power failure to the actuator, the electrical or pneumatic force is no longer present and the spring pressure forces the valve to open.

2.1. Problems associated with valves on board vessel.

On board, a vessel, the various valves located in the Engine and deck department as well as other parts of the ship are bound to mechanical faults which can result in malfunctioning of

the valves. Below are some highlighted problems with valves on board vessels. Some of these include:

1. Stiction. One common issue found in the control valve loop is Stiction, a combination of words stick and friction. This term is a condition that holds the valve in a particular condition and sticks it there, not allowing it to move like a Dead band. In a stiction condition, the movement of the valve needs additional force, which can also cause the valve to overshoot its position and its process setpoint like the Dead band, leading the valve to get stuck in a new or wrong position. Stiction is often found in control loops operating in pre-set mode as a continuous cycling pattern of controller output and a square wave pattern of process variables producing process disturbance and valve tears.

2. Positioner Overshoot. : The additional problem in these recent years found to be more common is position overshoot. These positioners are fast feedback controllers measuring the valve stem positioners that manipulate the valve actuator to reach the desired position of the valve. Most of the valve positioners are capable of tuning; they have tuned too aggressively for their particular controlling valves causing the valve to overshoot its position with a change in controller output.

3. Dead Band. Dead band is a situation where there is no output for any input; the control valve system is essentially dead. Dead band mainly occurs when the valves need to change the direction, either intentionally or accidentally, from sources. Rise in loop deadtime is the Dead band divided by the change of controller output rate. The peak and integrated errors can be increased in load disturbance due to an increase in dead time. In case of dual or more integrators in the system due to integral action in the valve positioner, uneven speed in controllers or processor limitations in cycles will develop.

4. Wrong Valve position indication. The main purpose of the hydraulic controlled valve is to operate it remotely. The ECR and CCR control panel for such valves are provided with position indicator showing the current position of the valve (Open or Shut). If the indicator is showing a wrong position, it may lead to confusion and operator may proceed not knowing the actual or real position of the valve. This is due to defective flow meter as well as defective pressure switch problem in indicator panel.

5. Positioner overshoot. The additional problem in these recent years found to be more common is position overshoot. These positioners are fast feedback controllers measuring the valve stem positioners that manipulate the valve actuator to reach the desired position of the valve. Most of the valve positioners are capable of tuning; they have tuned too aggressively for their particular controlling valves causing the valve to overshoot its position with a change in controller output.

6. Incorrect valve sizing. Furthermore, problems in control valves are the improper sizing or oversized valves. These incorrect-sized valves can make the problems worse, especially in

smaller-sized valves. Valves should be in the proper size to obtain the travel flow accurately, depending on the valves' characteristics, curves, and service conditions. An oversized control valve, even a slight change in valve position can affect the flow hugely, and in undersized control valves, there are situations of bottleneck occurring. In some cases, there are larger-sized control valves than the flow rate they need to control, leading to the valve operating at a small opening even at full flow conditions.

7. Non-linear flow characteristic. : Control valves are designed to have a linear flow. Hence, a non-linear flow also leads to tuning problems. So, all the above conditions including, Dead band, stiction, improper sizing, and positioner overshoot, can cause non-linearity. The flow characteristic of the control valve is the relationship between flow rate and valve position under normal operating conditions, which is supposed to be linear. A nonlinear characteristic can provide only optimal controller response at the operating point; this loop could be utterly unstoppable or stagnant as the valve position may move away from its operating point.

8. Valve does not shut entirely. A valve needs to shut fully as a partially closed valve may lead to continue in the flow and result in spilling of the fluid. This may be due to Internal leakage of hydraulic actuator, oil pressure too low, the problem in relief and check valve, the operation time of the valve is too slow.

9. Stationary valve. It may happen that a valve which is signaled from the CCR or ECR to open or shut is not moving at all. This may lead to a spill of fluid (if the valve is stuck in open position) or over□pressurization of the connected piping system (if the valve is stuck in a closed position). No movement of the valve is caused by problem in electric control system, problem in hydraulic actuators, valves or pipes.

10. Noisy relief valve which is caused by setting to too low or too close to another valve setting.

11. Worn out puppet and seat of the valve.

12. Worn and damaged pressure relief valve and improper adjustment which is caused by excessive pressure

CHAPTER THREE.

PROGRAMMABLE LOGIC CONTROLLERS (PLC).

Introduction to PLC (Programmable Logic Controllers).

3.0. What is a Programmable Logic Controller (PLC)?

A PLC (programmable logic controller) is an industrial digital computer designed for the control and automation of manufacturing processes or robotic devices. PLCs are capable of doing all monitor and control functions used to perform in the process control industry and have been ruggedized and adapted for the control of manufacturing processes, such as assembly lines of automobiles, FPSOs, Navy ships, machines, robotic devices, or any activity that requires high reliability, ease of programming, and process fault diagnosis. In comparison to the conventional relay circuit, PLC almost eliminated the heavy hardwiring problem when compared to relay circuits. In addition to the hardwiring problem, PLCs have more advantages over conventional relay-based systems

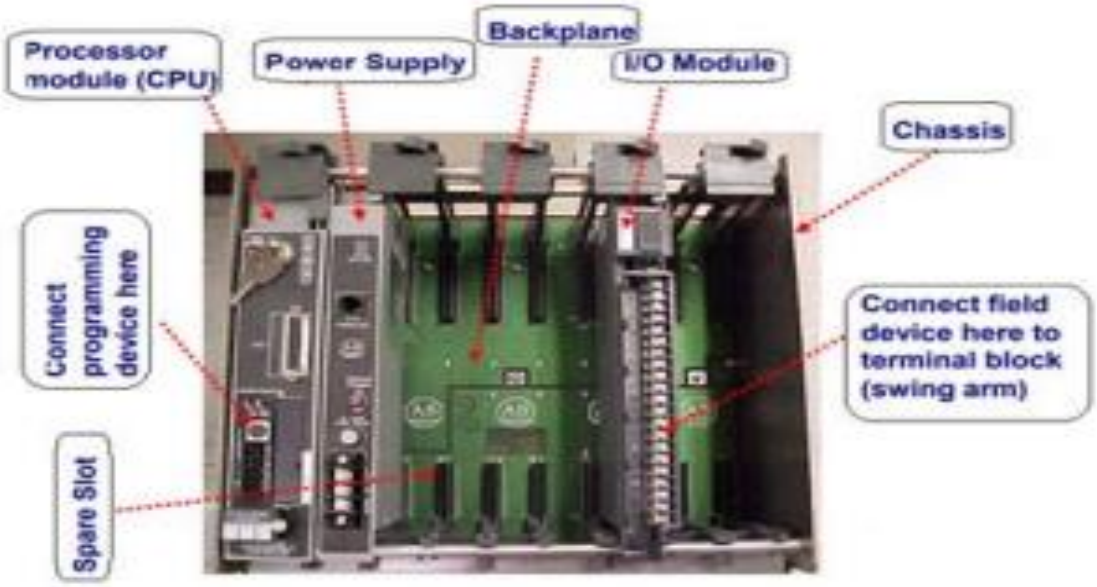
Dick Morley is considered as the father of PLC as he had invented the first PLC, the Modicon 084, for General Motors in 1968. PLCs can range from small modular devices with tens of inputs and outputs (I/O), in a housing integral with the processor, to large rack-mounted modular devices with thousands of I/O, and which are often networked to other PLC and SCADA systems. They can be designed for many arrangements of digital and analog I/O, extended temperature ranges, immunity to electrical noise, and resistance to vibration and impact. Programs to control machine operation are typically stored in battery-backed-up or non-volatile memory. PLCs were first developed in the automobile manufacturing industry to provide 14 flexible, rugged and easily programmable controllers to replace hard-wired relay logic systems. Since then, they have been widely adopted as high-reliability automation controllers suitable for harsh environments. A PLC is an example of a hard real-time system since output results must be produced in response to input conditions within a limited time, otherwise unintended operation will result

Architecture of programmable logic controllers.

A PLC is an industrial microprocessor-based controller with programmable memory used to store program instructions and various functions. It consists of:

1. a processor unit (CPU) which interprets inputs, executes the control program stored in memory and sends output signals,
2. a power supply unit which converts AC voltage to DC,
3. a memory unit storing data from inputs and program to be executed by the processor,

4. an input and output interface, where the controller receives and sends data from/to external devices,
5. a communications interface to receive and transmit data on communication networks from/to remote PLCs.



3.1. Types of PLCS.

Programmable Logic Controllers (PLCs) have become very crucial in defining and enhancing industrial production. Their robust construction that includes advanced features such as PID controllers and timers make it easy to program industrial systems for efficient production, easy 15 diagnostics, and safety. PLCs are integrated into production systems as single or module units. Some of these types based on size, power supply and hardware setup include; Modular PLCs, Compact PLCs, Hard and Soft PLCs, Unitary PLCs, Slot PLCs and many more used for industrial purposes. Listed below are a few highlights on some types of the PLCs and their functions.

PLC type based on hardware setup.

Integrated / Compact PLC. This type of PLC is constructed using a number of modules that are housed in the same case. This implies that the manufacturer, as opposed to the user, decides the Input/output (I/O) capabilities. Some integrated PLCs are designed to allow connection of addition I/O in order to look more like the module design

Modular PLC. The modular PLC is a type that allows multiple expansions of the PLC system through the use of modules, hence the term “modular”. Modules give the programmable logic controller additional features like increased number of I/O units, and they are usually easier to use because each component is independent of each other. The power supply, communications module, Input/Output module are all separate to the actual

microcontroller to manually connect them to each other to create the PLC control system. A type of modular PLC is the rack-mounted or rack mount PLC. In a rack mount PLC, the communications module of the PLC resides in the rack itself, so all connections are centralized

PLC types based on power supply.
PLCs are available with or without SMPS (Switch Mode Power Supply).
1. PLC with SMPS (230V AC)
2. PLC without SMPS (24V DC)
Some PLCs can be connected to a 230 VAC power supply directly to it. This means these have inbuilt SMPS which converts the supplied 230 VAC into the standard 24 VDC. Some PLCs needs a 24 V DC which doesn't have the inbuilt SMPS. In this case, we need an external switched-mode power supply (SMPS) for the conversion of 230 VAC into 24 VDC.

PLC types based on size.
The PLCs are further classified depends on the number of inputs and outputs a PLC can handle. The PLCs are classified based on size are as follows
1. Nano PLC.
2. Micro PLC.
3. Medium.
4. Large.
5. Very Large.
The memory capacity of these PLCs are also varies based on their I/O capacity.

3.2. How does a PLC operate?

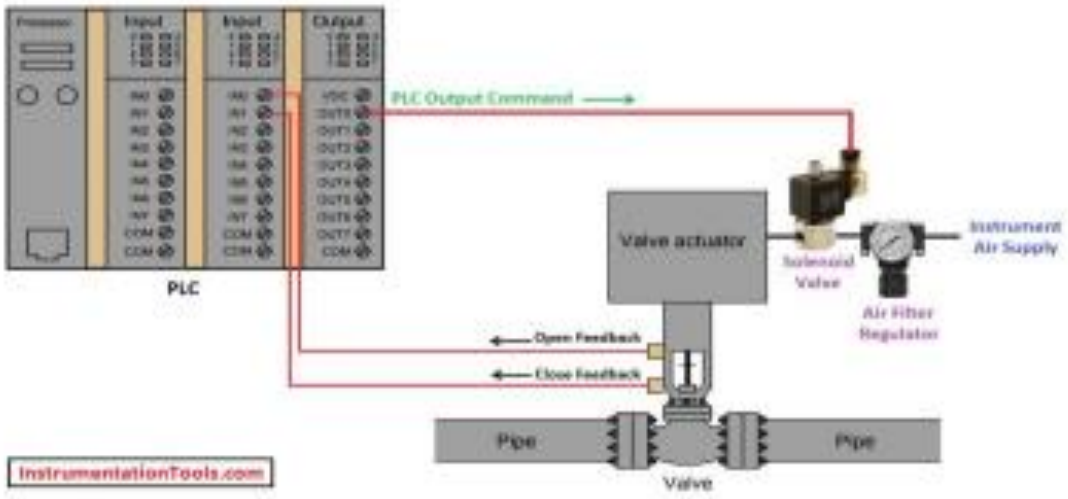
There are four basic steps in the operation of all PLCs; Input Scan, Program Scan, Output Scan, and Housekeeping. These steps continually take place in a repeating loop. The Four Steps in The PLC Operations include:

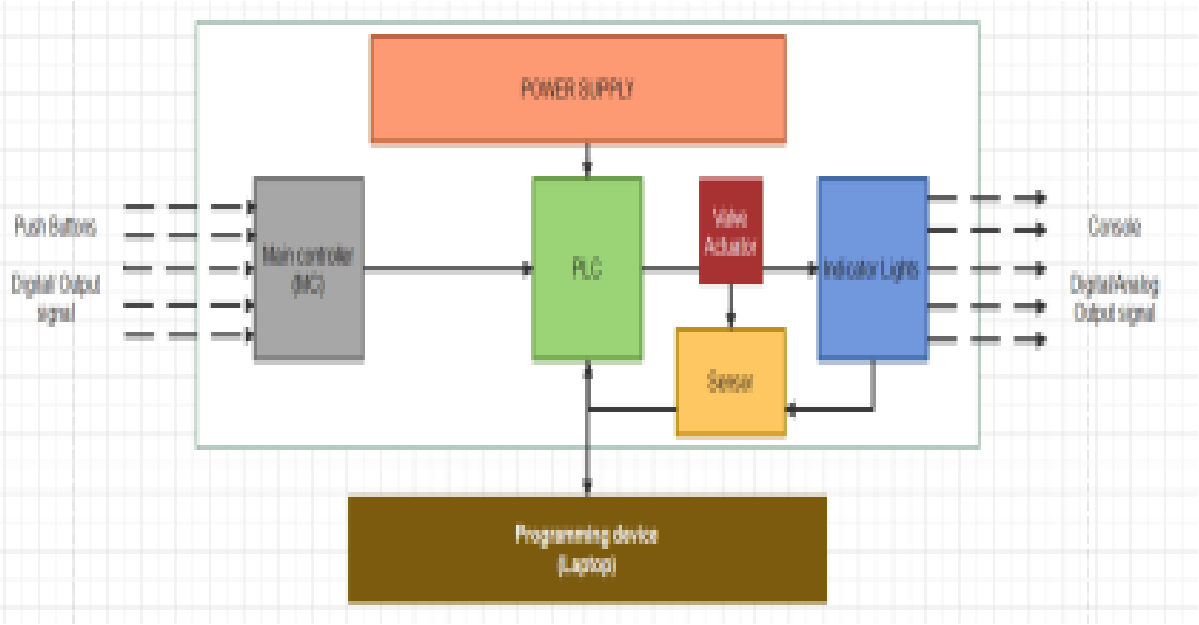
1.) INPUT SCAN. Detects the state of all input devices that are connected to the PLC
2.) PROGRAM SCAN. Executes the user created program logic
3.) OUTPUT SCAN. Energizes or de-energize all output devices that are connected to the PLC.
4.) HOUSEKEEPING.
This step includes communications with programming terminals, internal diagnostics, etc...

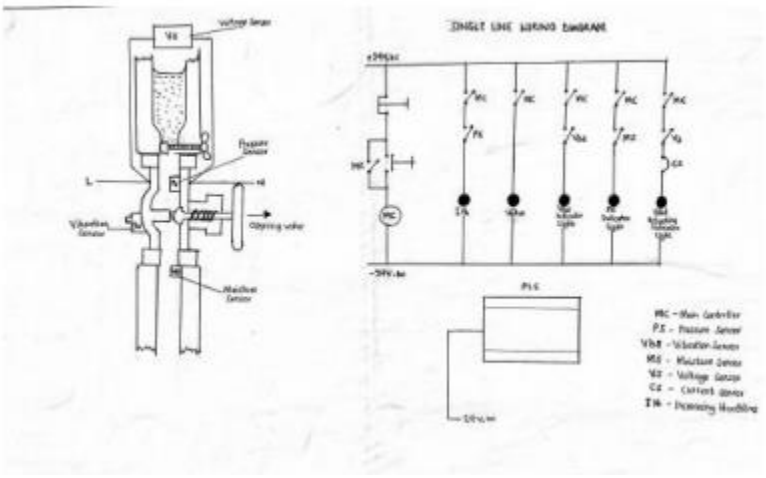
3.3. Using PLC to troubleshoot valve problems on board the vessel.

Although not all the problems associated with valves were addressed, a few of them as categorically stated in the report were addressed and further went on to understand how the PLC can help solve the problems. Some of these identified issues we adopted to solve were: Valve leakages, Jerking, excessive pressure, bad actuation, etc. The software used was the Siemens step 7 for the analysis. To resolve or troubleshoot valve issues on board a vessel using programmable logic controllers, the following components are required:

- A PLC with a 24 V DC SMPS.
- A main controller.
- Push buttons.
- Sensors. (5V)
- A PLC Console.
- Contactor.
- Motor.
- Power supply.





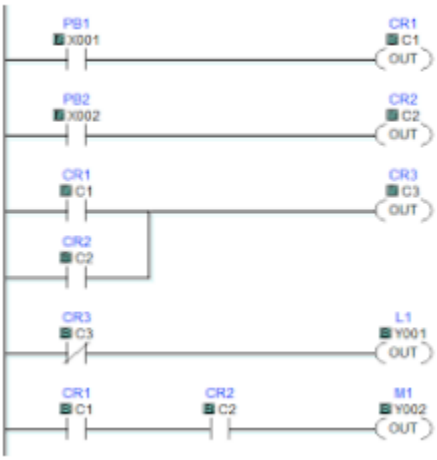


A single line wiring diagram was generated to help show the control circuit diagram of the setup as shown above

From the diagram above, the main power supply is connected to the PLC. A system is then developed such that a single line wiring diagram is generated and connected to the PLC. The starting mechanism controls the main controller. The main controller (MC) serves as a virtual circuit board where power can be supplied using the push buttons. Based on the identified problems, it was agreed that various sensors and indicator lights were installed and used in the circuit drawing to provide outputs that would be read from the PLC console to help identify the issue on board from the control room. Simulating this into a practical

approach with the valve connected to the main power source, the following scenarios were considered. A). For excessive pressure build-up, a throttling valve was used since PLCs can check flow rates. The throttling valve was then synchronized to the main controller to the main supply and then to the PLC. This is then connected to a pressure sensor and then to the pressure indicator such that for excess pressure or high pressures, the throttling valve is opened gradually until an optimal pressure is attained and then closes the circuit. This then helps adjust an optimal flow rate discharge through the valve to its designated output source. However, when there is no build-up pressure, the circuit is opened and no signal is sent to the console. 24 B). When the valve begins to jerk or due to too much vibration on the valve, we introduced a vibration sensor that will detect noise and vibrations and immediately send a signal to shut off the valve immediately so that it can be worked on. With these few highlighted scenarios, a code was written using the ladder logic programming language to help visualize the results obtained during the analysis.

The Ladder Logic Code



```
<Program>
  <Routine Name="FB 100" Type="Function Block(FB)">
  • • •
  <Routine Name="FB 421" Type="Function Block(FB)">
    <Rung Number="1">
    <Rung Number="2">
      <Output Type="MOVE" Source_Value="0" Target_Address="MW3755">
        <Condition Type="AND" Address="M1.5"/>
      <Output Type="MOVE" Source_Value="0" Target_Address="MD3650">
      <Output Type="MOVE" Source_Value="0" Target_Address="MW250">
      <Output Type="Output Coil" Address="Q215.0">
      <Output Type="MOVE" Source_Address="MW255" Target_Address="MW257">
      <Output Type="Output Coil" Address="M7131.6">
      <Output Type="Set Coil" Address="Q215.1">
      <Output Type="Convert BCD to Integer" Source_Address="MW2" Target_Address="MW2546">
      <Output Type="Add Integers" First_Input="MW0" Second_Input="MW7" Output_Adress="MW3955">
      <Output Type="Reset Coil" Address="Q215.2">
    <Rung Number="3">
  • • •
    <Rung Number="16">
  <Routine Name="FB 422" Type="Function Block(FB)">
  • • •
  <Routine Name="FC 490" Type="Function(FC)">
```

CHAPTER FOUR

CONCLUSION

As valves are considered as technical fittings in every industry and even on-board ships, it is important that they are maintained properly under good conditions and follow precautionary maintenance routines to ensure they are efficient and effective. As modern technology has advanced, PLCs are now integrated on most vessels to assist in controlling control elements as well as automate processes on board a ship. As a team, we were able to identify some problems that are associated with valves on vessels and we have come to realize that valves on vessel can cause flooding of compartments in the ship when the valve that controls the fluid flow leaks due to the corrosion of the valve caused by the sea water of high salt content. Difficulties in opening of valves and closing is also an issue that seafarers faces and some other valves problems as well. Using PLC which is a controller to control valves onboard, it gives an indication as a signal to the valve to execute. The controller compares the outcome to the set standard. If the outcome overweighs the set standard, the controller indicates and send signal for the valve to control the outcome before it executes. In future, PLC to control valves, there is the need to solve the problem on how it turns on when power is been restored.

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